

Solution Architecture

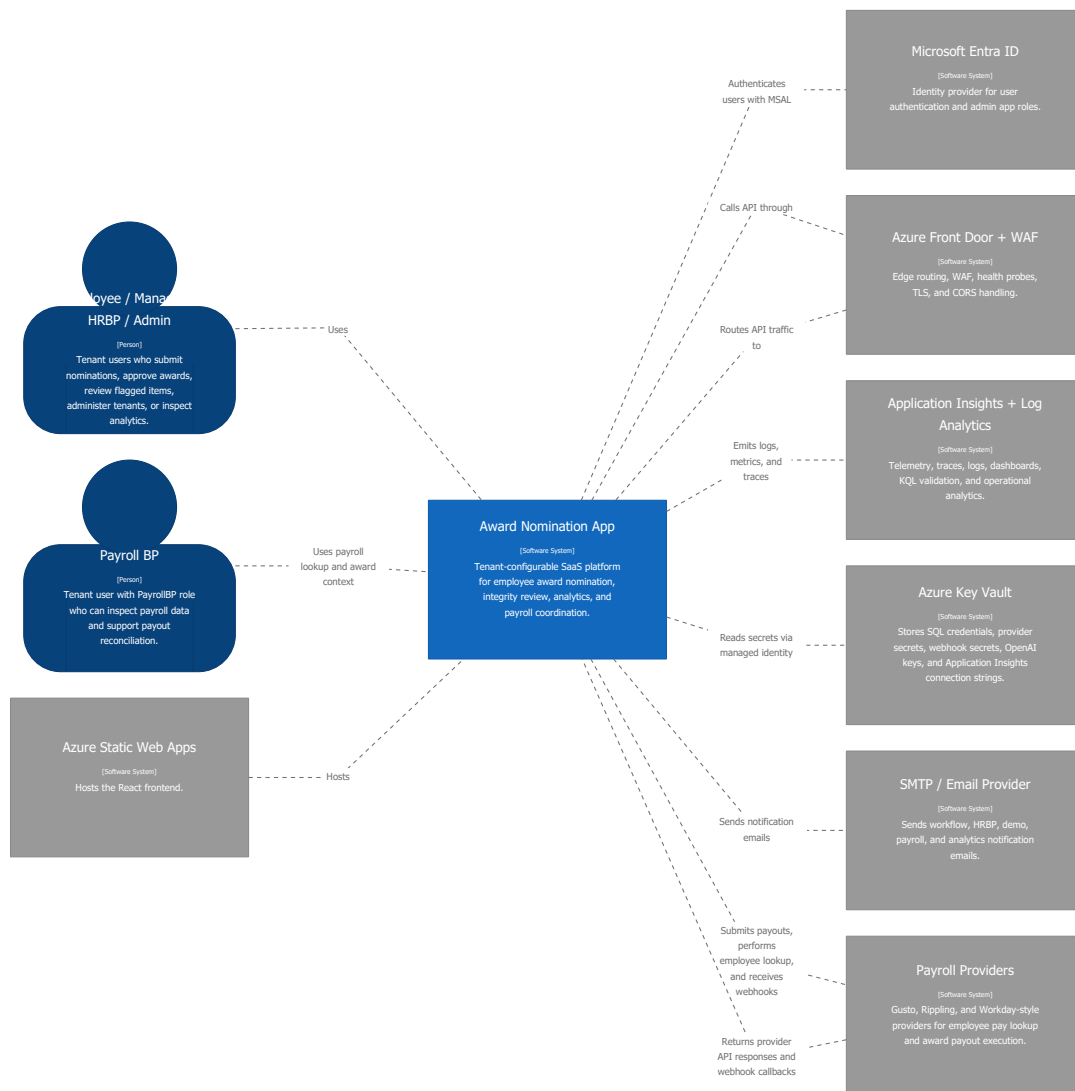
Architecture Summary

The solution is an Azure-native SaaS application with:

- React/Vite frontend hosted by Azure Static Web Apps.
- FastAPI backend hosted on Azure Container Apps in primary and secondary regions.
- Azure Front Door Standard and WAF routing external traffic to backend origins and payroll broker origins.
- Azure SQL as the relational system of record.
- Azure Service Bus topic and subscriptions for asynchronous workflow events.
- Separate worker services for integrity checking, notifications, payout handling, and payroll processing.
- Azure Blob Storage for fraud models, certificates, templates, and exports.
- Azure OpenAI and local ML libraries for analytics, semantic checks, and investigation support.
- Application Insights and Log Analytics for observability.

System Context

Structured source: `../diagrams/structurizr/workspace.dsl`, view `SystemContext`.

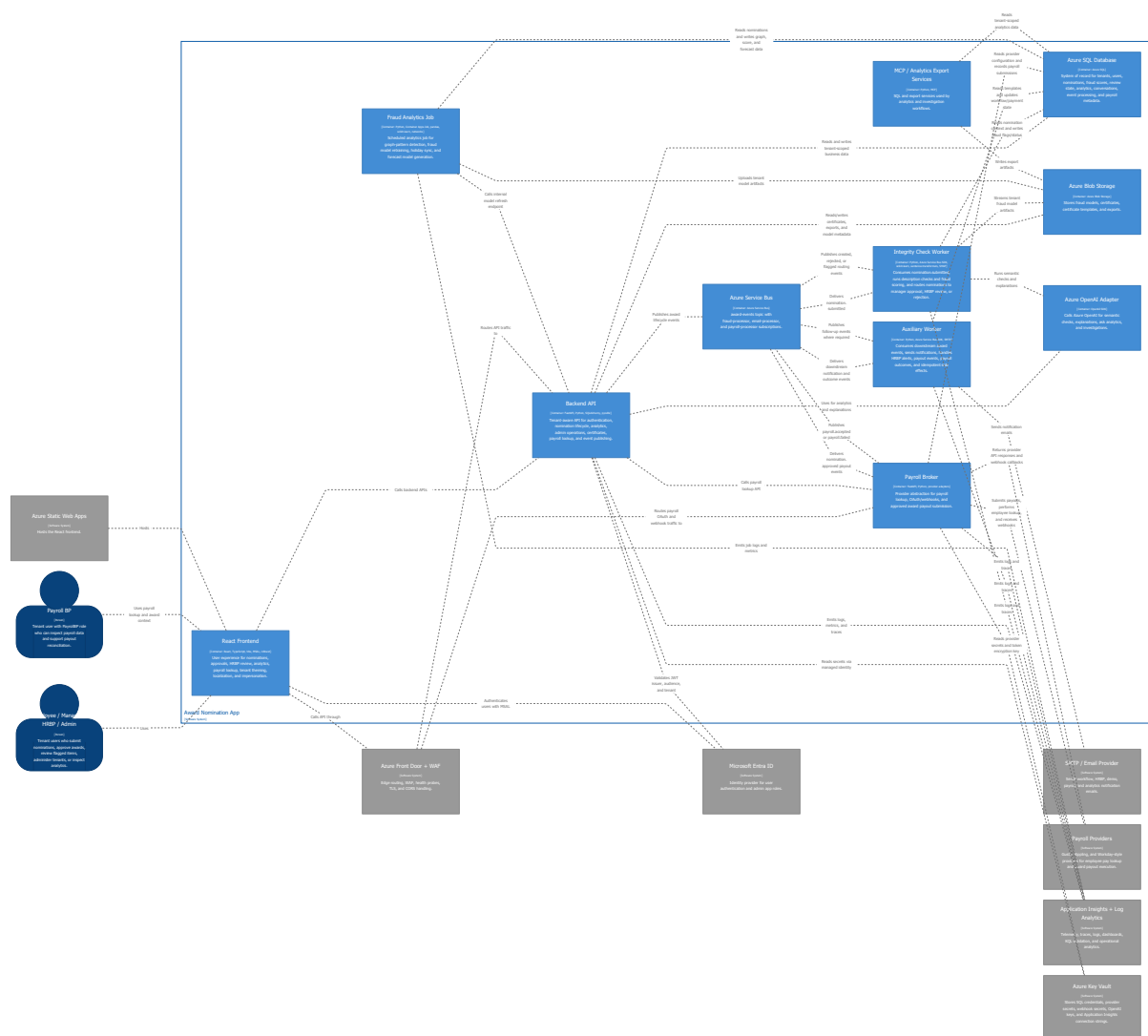


System Context View: Award Nomination App
System context for users, identity, Azure services, payroll providers, and observability.
Tuesday, July 7, 2026 at 3:48 PM Pacific Daylight Time

System context

Major Containers

Structured source: [../diagrams/structurizr/workspace.dsl](#), view [ContainerArchitecture](#).



Container architecture

Container	Path	Primary responsibility
Frontend	<code>frontend</code>	User experience, MSAL sign-in, tenant theming, nomination UI, approvals, analytics, HRBP, payroll lookup.
Backend API	<code>backend</code>	Authentication, tenant resolution, nomination APIs, analytics APIs, admin functions, certificate SAS links, Service Bus publishing.
Integrity Check Worker	<code>integrity-check</code>	Consumes <code>nomination.submitted</code> , runs description checks and fraud assessment, routes clean/flagged/rejected outcomes.
Auxiliary Service	<code>auxiliary-service</code>	Consumes downstream events, sends email, handles HRBP alerts, payout submission, payroll results, idempotency.
Payroll Broker	<code>payroll-broker</code>	Consumes approval events, routes payout to provider, exposes OAuth/webhook endpoints, supports pay lookup.
Fraud Analytics Job	<code>fraud-analytics-job</code>	Runs weekly graph detection, model training, holiday sync, and forecast model generation.
MCP / Analytics services	<code>MCP_Servers</code>	SQL and export service support for analytics/investigation workflows.

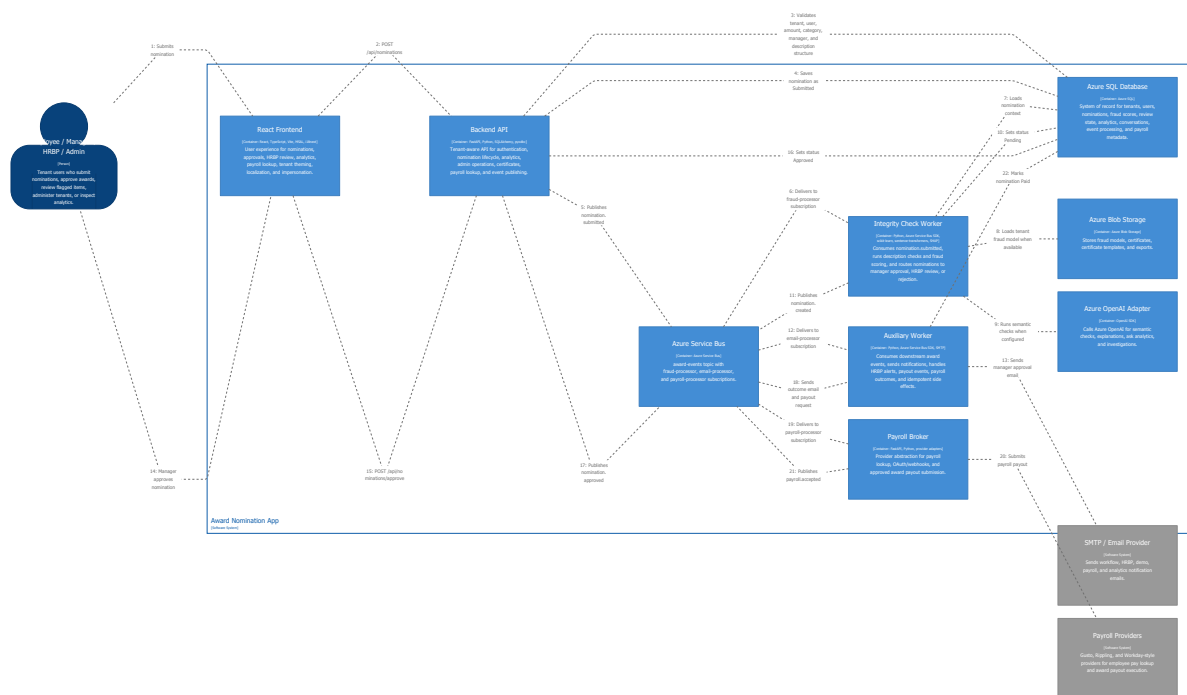
Tenant Boundary

The tenant boundary is enforced through:

- Entra ID **tid** claim.
- **Tenants.AzureAdTenantId** allowlist lookup.
- Internal **TenantId** attached to the authenticated user context.
- Tenant-scoped user lookup.
- Tenant-scoped SQL helper functions and analytics queries.
- Domain-based tenant branding and wrong-portal detection.
- Tenant-specific configuration and categories.

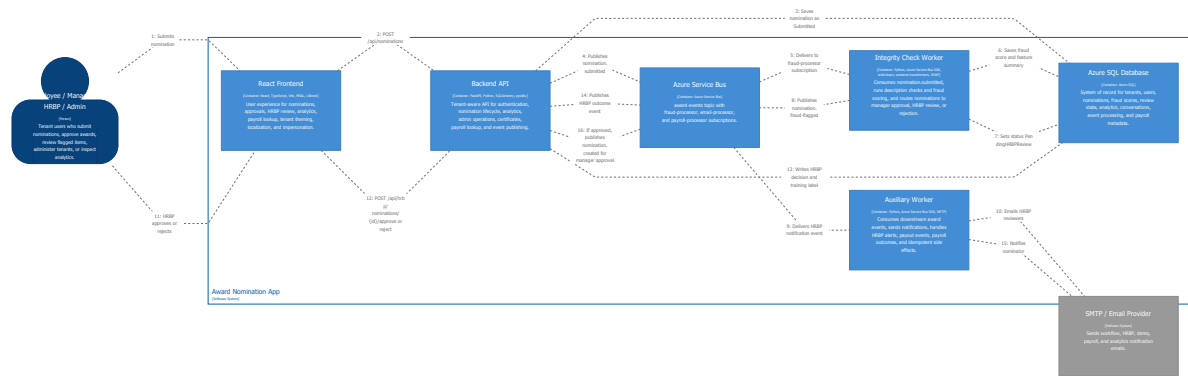
Core Workflow

Structured source: **../diagrams/structurizr/workspace.dsl**, views **NominationCleanApprovalFlow** and **NominationHRBPReviewFlow**.



Dynamic View: Award Nomination App
Clean nomination path from submission through manager approval and payroll submission.
Tuesday, July 17, 2024 at 3:44 PM Pacific Daylight Time

Clean nomination approval flow



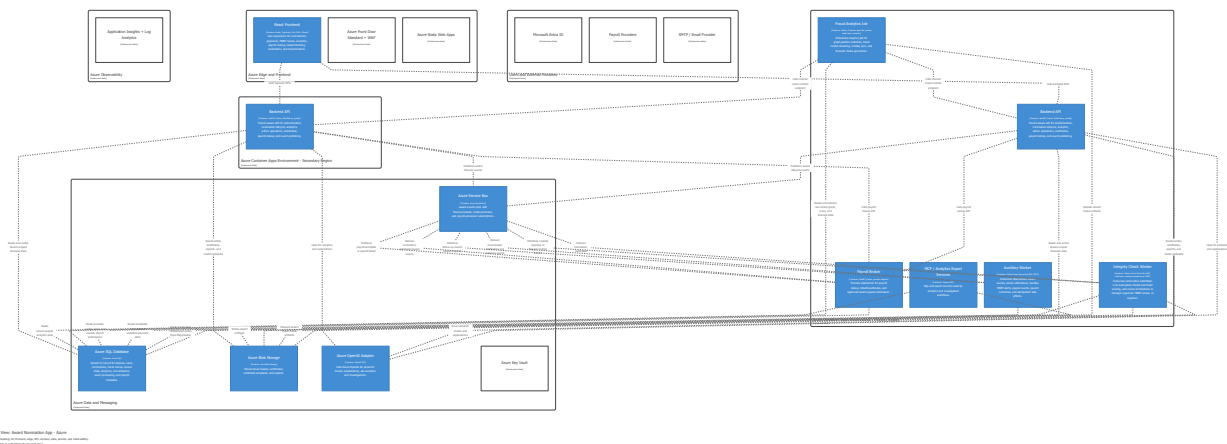
HRBP review nomination flow

Service Bus Topology

Topic or subscription	Filter	Consumer
award-events topic	All award lifecycle events	All publishers send here.
fraud-processor subscription	event_type = 'nomination.submitted'	Integrity Check Worker.
email-processor subscription	event_type != 'nomination.submitted'	Auxiliary Service.
payroll-processor subscription	event_type = 'nomination.approved'	Payroll Broker.

Deployment Topology

Structured source: `../diagrams/structurizr/workspace.dsl`, view `AzureDeployment`.



External Dependencies

- Microsoft Entra ID for authentication and app roles.
- Azure SQL for transactional and analytics persistence.
- Azure Service Bus for workflow events.
- Azure Blob Storage for models, certificates, certificate templates, and exports.

- Azure OpenAI for AI analytics, investigation, and explanation generation.
- SMTP/email provider for workflow notifications.
- Gusto/Rippling/Workday-style payroll providers for payout execution.
- Cloudflare/Azure DNS assets for public tenant and payroll broker domains.